A CONSTANT-KERNEL DIAGONAL COUNTERMODEL TO A HEAT-TRACE IMPLICATION FOR ERDŐS PROBLEM #1055

ARGUS OPEN RESEARCH PROJECT

Abstract. We exhibit an elementary all-N diagonal matrix family $H_N = \mathrm{diag}(0_K, (3/10)I_{N-K})$ showing that the bounded-kernel heat-trace inequalities in a 2025 manuscript on Erdős Problem #1055 and Cunningham chains do not imply an unbounded kernel dimension as $N \to \infty$.

1. The Countermodel

Let $K \geq 1$ be a fixed integer, and for every $N \geq K$ define the $N \times N$ positive semidefinite diagonal operator

$$H_N \;=\; \mathrm{diag}\big(\underbrace{0,\dots,0}_{K},\ \underbrace{\tfrac{3}{10},\dots,\tfrac{3}{10}}_{N-K}\big).$$

Proposition 1.1. *For every fixed $K \geq 1$ and every $N \geq K$:*

(1) $\dim\ker(H_N) = K$ *is constant and bounded by* $C = K$.

(2) *The smallest positive eigenvalue of* H_N *is* $\lambda_1(H_N) = 3/10 > 0$.

(3) *For every* $t > 0$, *the heat trace satisfies*

$$(N-C)e^{-3t/10} \;\leq\; \mathrm{Tr}(e^{-tH_N}) \;=\; K + (N-K)e^{-3t/10} \;\leq\; C + (N-C)e^{-3t/10}.$$

The upper and lower bounds in (3) differ by $C = K > 0$; *equating them in the limit* $N \to \infty$ *without dividing by* N *is invalid, and the inequalities are satisfied by a family with constant kernel dimension* K.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2457 in the domain of spectral theory and analytic number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *proof premise countermodel* deciding the scope: heat-trace bounded-kernel implication in viXra:2506.0023v1 refuted by H_N=diag(0_K,(3/10)I_{N-K}).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- Erdos Problem #1055 on Cunningham chains.

References

[1] Primary problem list and source literature for Result 2457 (spectral theory and analytic number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 11A41; Secondary 15A18.